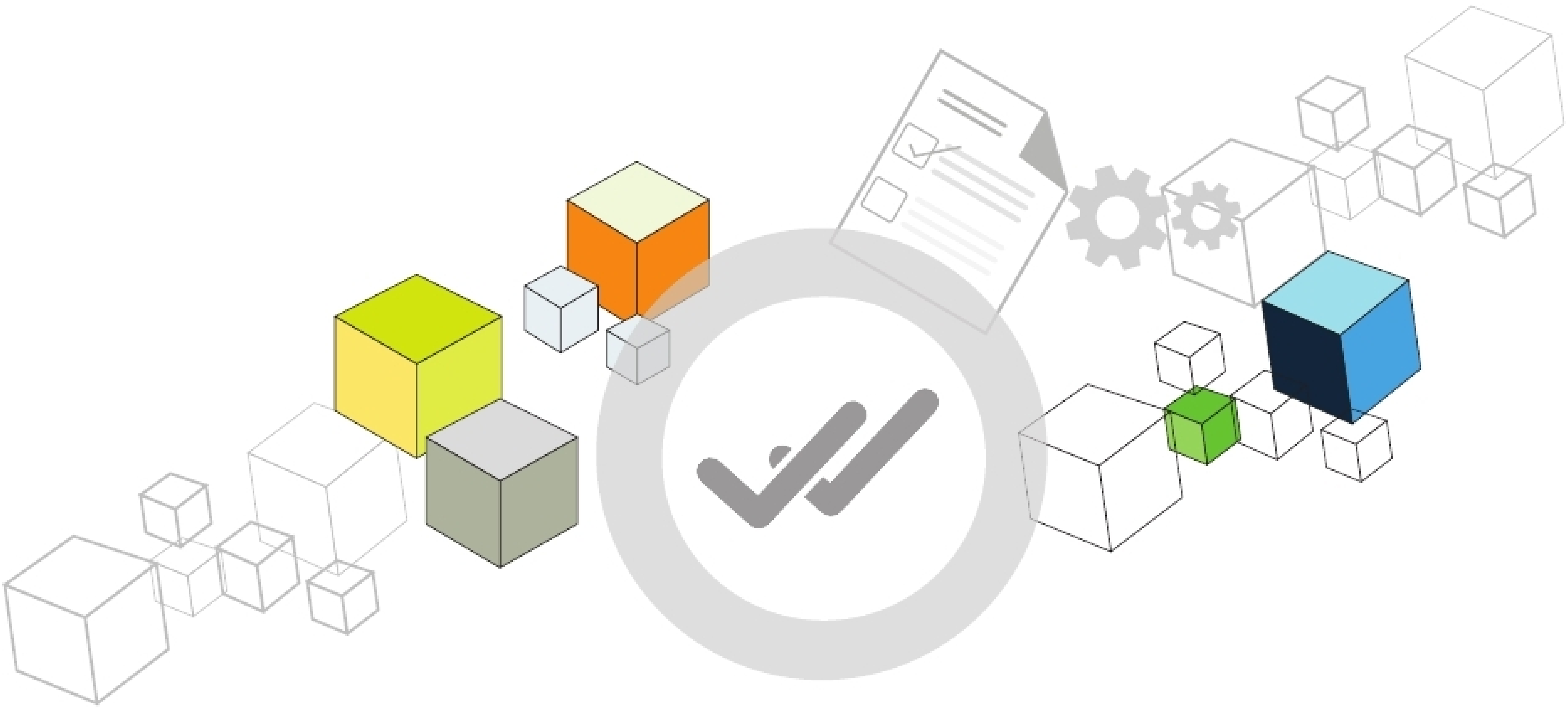
It's Important to Validate Blockchain Applications in Real-time Scenarios

Cryptocurrencies and their underlying blockchain based technology have created a buzz over the past few years. In this article, we take a look at how we can test blockchain based applications.



A blockchain is a decentralised, tamper-proof database that has been called the trust protocol of the Internet. It helps in settling transactions between multiple anonymous entities over the Internet without going through a centralised service. The transactions in a blockchain are settled in a decentralised manner using a consensus mechanism, which is unique to the network, without the interference of any central authority. This ensures that the transactions are securely only under the user's control, throughout the entire process. Moreover, this enables individuals or enterprises to leverage the power of the blockchain to manage and access their information, and ensures trust on the network. Bitcoin is the current market leader and the largest blockchain network, followed by Ethereum, Ripple, Bitcoin Cash, Litecoin, etc. The list is huge and is growing on a constant basis.

Typical blockchain usage and applications

Here are some major enterprise applications of the blockchain:

- In the financial services sector as a replacement for fiat currencies, for payment processing, to build a transparent system for borrowing and lending
- In the manufacturing sector for supply chain management and legal contracts
- In the energies and utilities sector for energy transactions between sellers and borrowers, and for payment processing
- In the real estate management sector for property transactions and legal agreements

Figure 1 depicts the general architecture of blockchain based systems and applications.

Validations that are required for blockchain applications

Blockchain adoption comes with its inherent challenges – high initial investments, integration problems with legacy systems, privacy and security issues, and apprehensions around the adoption process. Hence, testing blockchain apps becomes business-critical.

There are multiple platform vendors who offer consultation services for blockchain implementation. You need to understand and decide what really suits your business needs and syncs well with your objectives.

Given below are the types of validations that can be used to ensure a high level of test coverage and quality for blockchain applications:

- Unit testing
- Integration testing
- User interface testing